

Project Setup Instructions

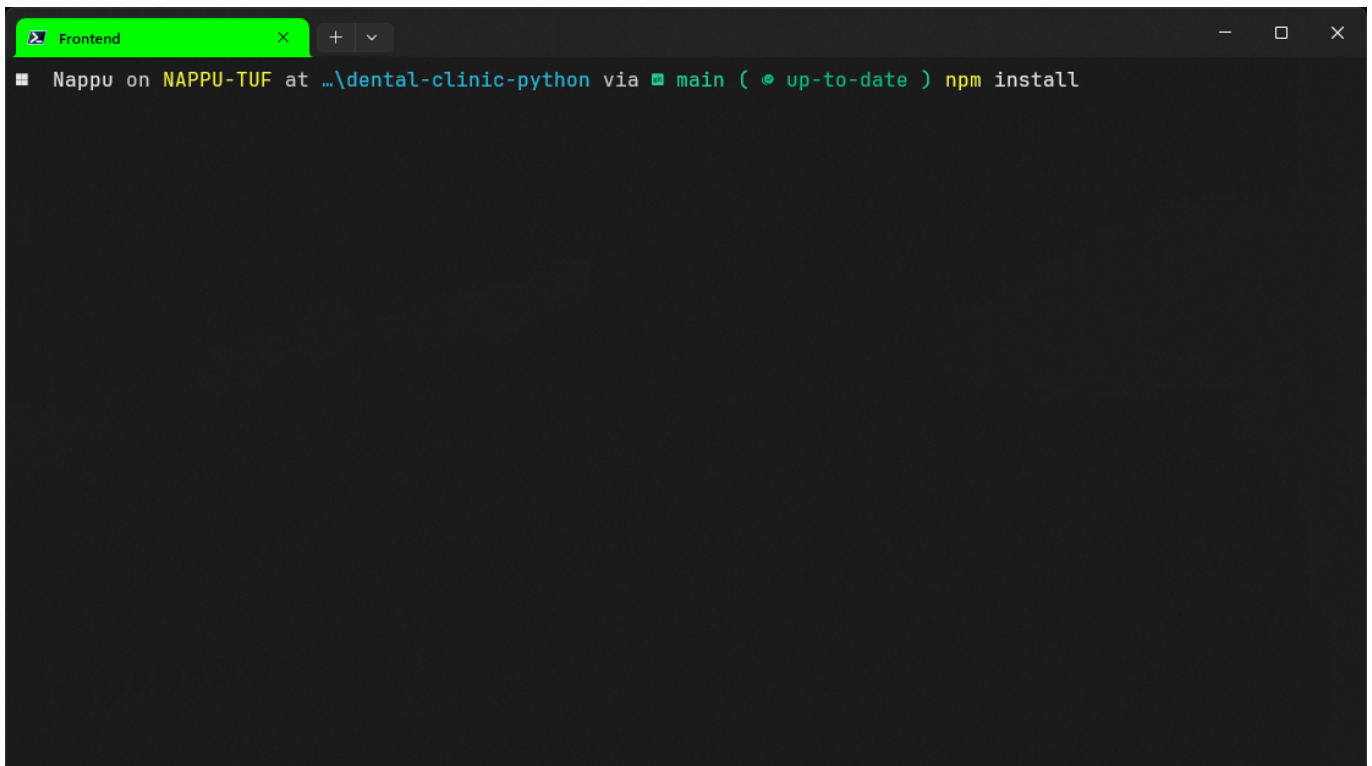
This document provides step-by-step instructions for setting up and running the frontend and backend development servers.

Frontend Setup

Step 1: Install Dependencies

Install all required npm packages for the frontend application.

```
npm install
```



Step 2: Start Development Server

Run the frontend development server.

```
npm run dev
```

```
Frontend x + v
■ Nappu on NAPPU-TUF at ...\\dental-clinic-python via ■ main ( ● up-to-date ) npm install
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out
lru-cache if you want a good and tested way to coalesce async requests by a key value, which is much more c
omprehensive and powerful.
npm warn deprecated rimraf@2.7.1: Rimraf versions prior to v4 are no longer supported
npm warn deprecated glob@7.2.3: Glob versions prior to v9 are no longer supported

added 386 packages, and audited 387 packages in 17s

57 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
■ Nappu on NAPPU-TUF at ...\\dental-clinic-python via ■ main ( ● up-to-date ) npm run dev
```

Step 3: Server Status

Verify that the frontend server is running successfully. Server should be running at <http://localhost:3000>

```
Frontend x + v
■ Nappu on NAPPU-TUF at ...\\dental-clinic-python via ■ main ( ● up-to-date ) npm install
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found 0 vulnerabilities
■ Nappu on NAPPU-TUF at ...\\dental-clinic-python via ■ main ( ● up-to-date ) npm run dev

> nextn@0.1.0 dev
> next dev

  ▲ Next.js 15.3.2
  - Local:      http://localhost:3000
  - Network:    http://192.168.100.13:3000

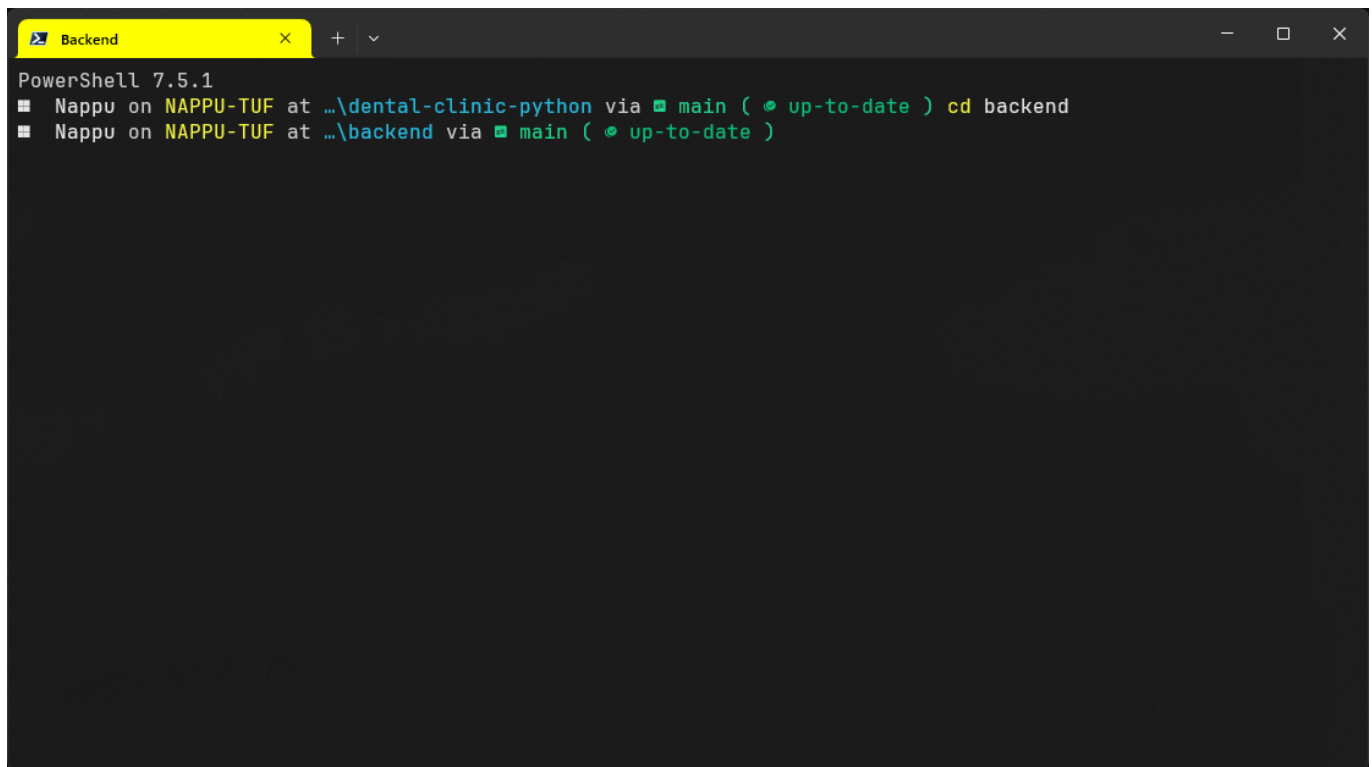
✓ Starting...
✓ Ready in 2s
```

Backend Setup

Step 4: Navigate to Backend Directory

Change to the backend directory to begin server setup.

```
cd backend
```



The screenshot shows a PowerShell terminal window with a yellow title bar labeled 'Backend'. The terminal output shows the user 'Nappu' on a system named 'NAPPU-TUF'. The first command is 'cd backend' executed from the directory '...\dental-clinic-python'. The second command is not fully visible but appears to be 'cd backend' executed from '...\backend'. The terminal background is dark, and the text is in a light color.

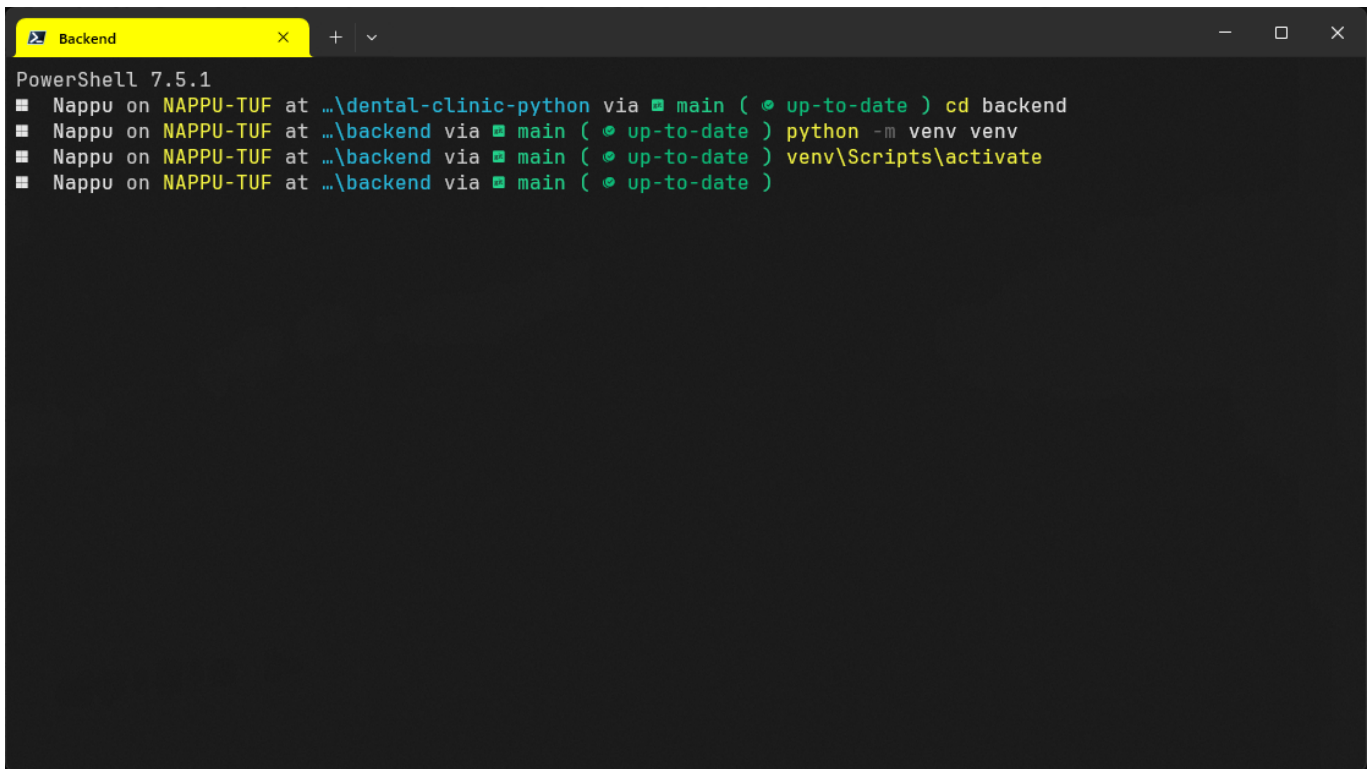
```
PowerShell 7.5.1
■ Nappu on NAPPU-TUF at ...\\dental-clinic-python via ■ main ( ● up-to-date ) cd backend
■ Nappu on NAPPU-TUF at ...\\backend via ■ main ( ● up-to-date )
```

Step 5: Python Virtual Environment

Create and activate a Python virtual environment for isolated package management.

```
# Windows
python -m venv venv
.\venv\Scripts\activate

# Linux/Mac
python3 -m venv venv
source venv/bin/activate
```

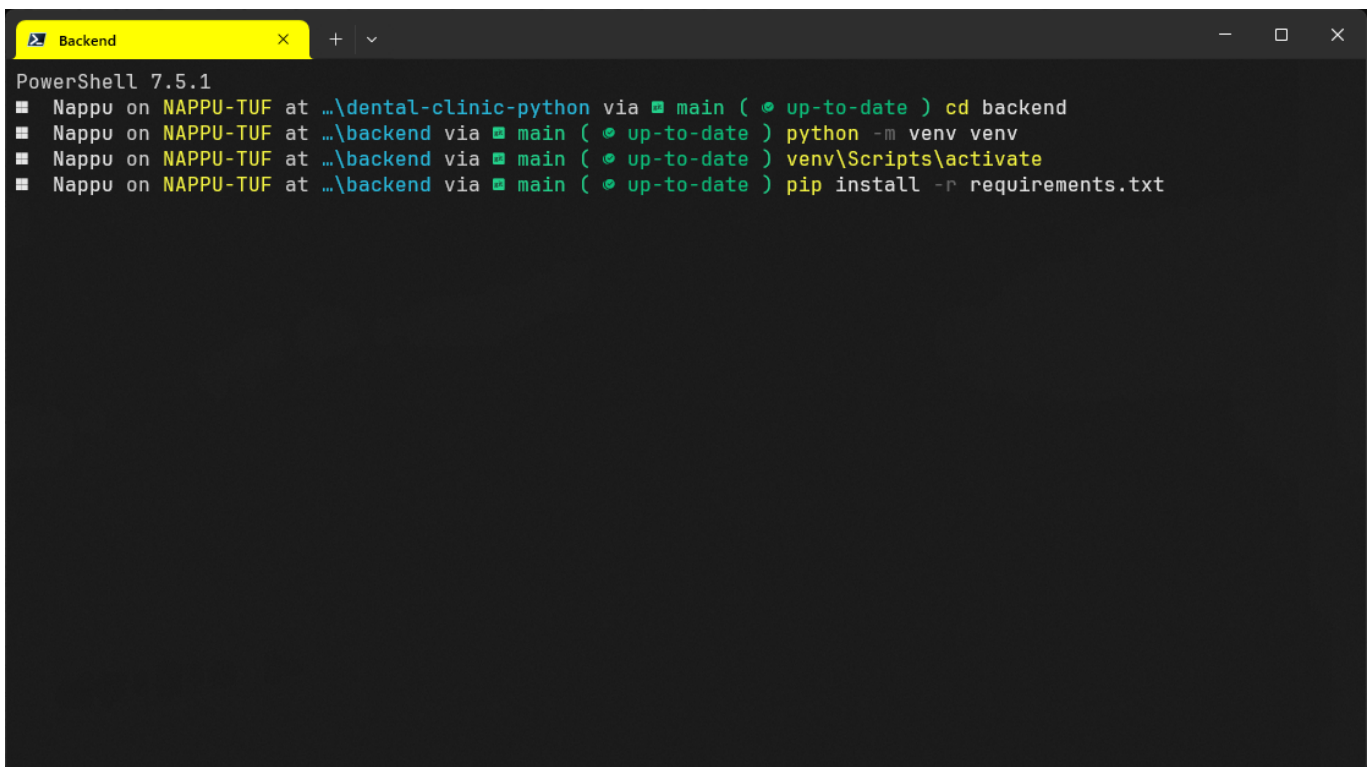


```
Backend x + v
PowerShell 7.5.1
Nappu on NAPPU-TUF at ...\\dental-clinic-python via main ( up-to-date ) cd backend
Nappu on NAPPU-TUF at ...\\backend via main ( up-to-date ) python -m venv venv
Nappu on NAPPU-TUF at ...\\backend via main ( up-to-date ) venv\\Scripts\\activate
Nappu on NAPPU-TUF at ...\\backend via main ( up-to-date )
```

Step 6: Install Requirements

Install the required Python packages from requirements.txt.

```
pip install -r requirements.txt
```



```
Backend x + v
PowerShell 7.5.1
Nappu on NAPPU-TUF at ...\\dental-clinic-python via main ( up-to-date ) cd backend
Nappu on NAPPU-TUF at ...\\backend via main ( up-to-date ) python -m venv venv
Nappu on NAPPU-TUF at ...\\backend via main ( up-to-date ) venv\\Scripts\\activate
Nappu on NAPPU-TUF at ...\\backend via main ( up-to-date ) pip install -r requirements.txt
```

Step 7: Package Installation Progress

Wait for all packages to complete installation.

```
Backend
PowerShell 7.5.1
Nappu on NAPPU-TUF at ...\\dental-clinic-python via main ( up-to-date ) cd backend
Nappu on NAPPU-TUF at ...\\backend via main ( up-to-date ) python -m venv venv
Nappu on NAPPU-TUF at ...\\backend via main ( up-to-date ) venv\\Scripts\\activate
Nappu on NAPPU-TUF at ...\\backend via main ( up-to-date ) pip install -r requirements.txt
Collecting fastapi>=0.100.0 (from -r requirements.txt (line 1))
  Using cached fastapi-0.115.12-py3-none-any.whl.metadata (27 kB)
Collecting uvicorn>=0.20.0 (from uvicorn[standard]>=0.20.0->-r requirements.txt (line 2))
  Using cached uvicorn-0.34.2-py3-none-any.whl.metadata (6.5 kB)
Collecting sqlalchemy>=2.0.0 (from -r requirements.txt (line 3))
  Using cached sqlalchemy-2.0.41-cp313-cp313-win_amd64.whl.metadata (9.8 kB)
Collecting pydantic>=2.0.0 (from -r requirements.txt (line 4))
  Using cached pydantic-2.11.4-py3-none-any.whl.metadata (66 kB)
Collecting passlib>=1.7.4 (from passlib[bcrypt]>=1.7.4->-r requirements.txt (line 5))
  Using cached passlib-1.7.4-py2.py3-none-any.whl.metadata (1.7 kB)
Collecting python-multipart>=0.0.5 (from -r requirements.txt (line 6))
  Using cached python_multipart-0.0.20-py3-none-any.whl.metadata (1.8 kB)
Collecting pytest>=7.0.0 (from -r requirements.txt (line 7))
  Using cached pytest-8.3.5-py3-none-any.whl.metadata (7.6 kB)
Collecting requests>=2.25.0 (from -r requirements.txt (line 8))
  Using cached requests-2.32.3-py3-none-any.whl.metadata (4.6 kB)
Collecting Faker>=15.0.0 (from -r requirements.txt (line 9))
  Using cached faker-37.3.0-py3-none-any.whl.metadata (15 kB)
Collecting httpx>=0.24.0 (from -r requirements.txt (line 10))
  Using cached httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting bcrypt=4.0.1 (from -r requirements.txt (line 12))
  Using cached bcrypt-4.0.1-cp36-abi3-win_amd64.whl.metadata (9.0 kB)
```

Step 8: Initialize Database

Set up the database schema and structure.

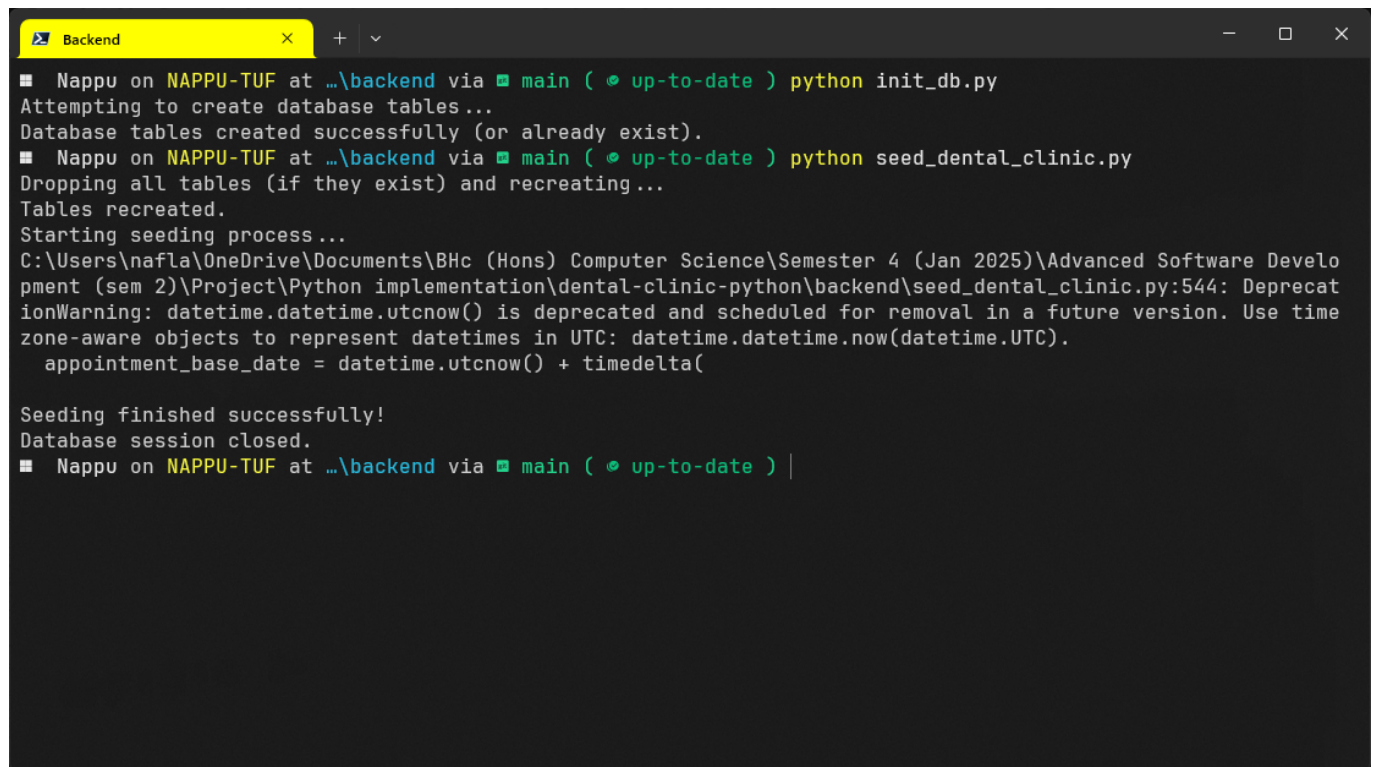
```
python init_db.py
```

```
Backend
Nappu on NAPPU-TUF at ...\\backend via main ( up-to-date ) python init_db.py
```

Step 9: Seed Database (Optional)

Populate the database with initial data if needed.

```
python seed_dental_clinic.py
```



The screenshot shows a terminal window titled 'Backend' with the following output:

```
■ Nappu on NAPPU-TUF at ...\\backend via ■ main ( ● up-to-date ) python init_db.py
Attempting to create database tables...
Database tables created successfully (or already exist).
■ Nappu on NAPPU-TUF at ...\\backend via ■ main ( ● up-to-date ) python seed_dental_clinic.py
Dropping all tables (if they exist) and recreating...
Tables recreated.
Starting seeding process...
C:\\Users\\nafla\\OneDrive\\Documents\\BHc (Hons) Computer Science\\Semester 4 (Jan 2025)\\Advanced Software Development (sem 2)\\Project\\Python implementation\\dental-clinic-python\\backend\\seed_dental_clinic.py:544: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use time zone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    appointment_base_date = datetime.utcnow() + timedelta(

Seeding finished successfully!
Database session closed.
■ Nappu on NAPPU-TUF at ...\\backend via ■ main ( ● up-to-date ) |
```

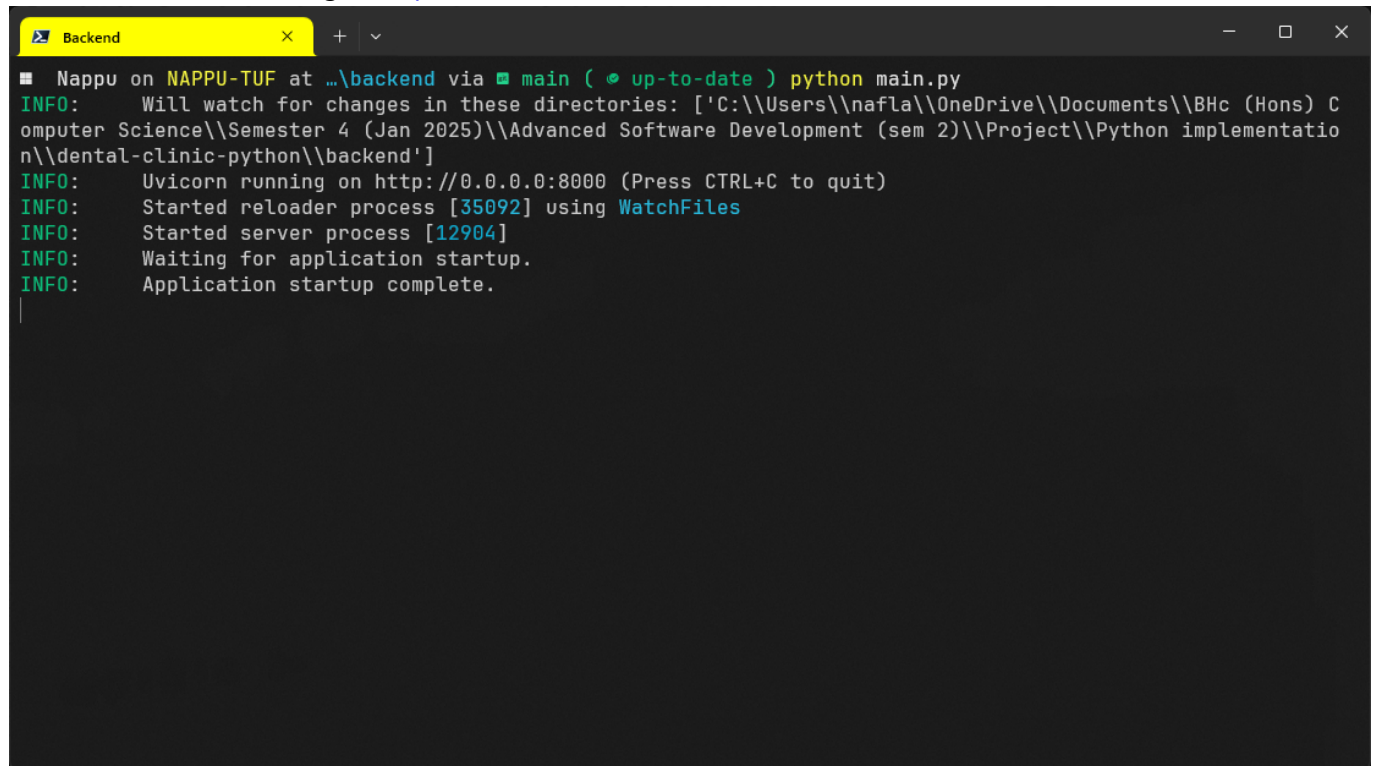
Note: You can skip Step 8 and 9 if the database file (`dev.db`) already exists in your backend directory.

Step 10: Start Backend Server

Launch the backend server using Python.

```
python main.py
# Server will start at
```


Server will be start running at <http://localhost:8000>



```
Backend x + v
■ Nappu on NAPPU-TUF at ...\\backend via ■ main ( ● up-to-date ) python main.py
INFO: Will watch for changes in these directories: ['C:\\Users\\nafila\\OneDrive\\Documents\\BHC (Hons) Computer Science\\Semester 4 (Jan 2025)\\Advanced Software Development (sem 2)\\Project\\Python implementation\\dental-clinic-python\\backend']
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: Started reloader process [35092] using WatchFiles
INFO: Started server process [12904]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Note: Make sure to follow these steps in order. The frontend and backend servers must both be running for the application to work properly. The database seeding step (Step 9) is optional and can be skipped if you don't need initial data. Access the application through Frontend Server

<http://localhost:3000>